

Zack Andalman

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EDUCATION

Princeton University	Princeton, NJ	Ph.D., Astrophysics	4.00 GPA	2023 - 2028
Yale University	New Haven, CT	B.S., Physics	3.95 GPA	2019 - 2023
Evanston Township HS	Evanston, IL	Diploma	4.00 GPA	2015 - 2019

PUBLICATIONS

First-Author

Published: 3

- [1] **Andalman, Z. L.**; Quataert, E.; Coughlin, E.; & Nixon, C. (2026)
Resolving the (Debate About) Nozzle Shocks in Tidal Disruption Events. OJAp.
<https://ui.adsabs.harvard.edu/abs/2026OJAp...962785A/abstract>
- [2] **Andalman, Z. L.**, Teyssier, R., & Avishai, D. (2025).
On the Origin of the High Star-Formation Efficiency in Massive Galaxies at Cosmic Dawn. MNRAS.
<https://ui.adsabs.harvard.edu/abs/2025MNRAS.540.3350A/abstract>
- [3] **Andalman, Z. L.**; Liska, M. T. P.; Tchekhovskoy, A.; Coughlin, E. R.; & Stone, N. (2022).
Tidal Disruption Discs Formed and Fed by Stream-stream and Stream-disc Interactions in Global GRHD Simulations. MNRAS.
<https://ui.adsabs.harvard.edu/abs/2022MNRAS.510.1627A/abstract>

Submitted: 1

- [4] **Andalman, Z. L.**; Fryer, C.; Fontes, C.; Mumpower, M.; & Wolleager, R. (2026).
Beta-Particle Transport and Thermalization in Kilonova Ejecta with Detailed Atomic Microphysics. MNRAS. <https://ui.adsabs.harvard.edu/abs/2026arXiv260701228A/abstract>

In prep: 1

- [5] **Andalman, Z. L.** & Quataert, E. (2026).
Limit Cycle and Stream Destruction: A Two-Stage Model for Circularization in Tidal Disruption Events. Planned submission to ApJL.

Co-Author

Published: 2

- [6] Sunseri, J.; **Andalman, Z. L.**; Teyssier, R. (2026).
Supermassive Black Hole Growth in Massive Galaxies at Cosmic Dawn. MNRAS.
<https://ui.adsabs.harvard.edu/abs/2026MNRAS.550g1129S/abstract>
- [7] Kaaz, N.; et. al. incl. **Andalman, Z. L.** (2023).
Nozzle Shocks, Disk Tearing and Streamers Drive Rapid Accretion in 3D GRMHD Simulations of Warped Thin Disks. MNRAS.
<https://ui.adsabs.harvard.edu/abs/2023ApJ...955...72K/abstract>

Submitted: 2

- [8] Yao, Z.; et. al. incl. **Andalman, Z. L.** (2026).
The MAGE Project: Size Evolution and Disk Formation of Massive Galaxies at Cosmic Dawn. MNRAS.

- [9] Nixon, C.; Coughlin, E.; & **Andalman, Z. L.** (2026).
On the Origin of Anomalous Dissipation in Simulations of Tidal Disruption Events. ApJ.
<https://ui.adsabs.harvard.edu/abs/2026arXiv260520327N/abstract>

In prep: 1

- [10] Jairam, P.; et. al. incl. **Andalman, Z. L.** (2026).
End-to-end Simulation of a Jetted Tidal Disruption Event. Planned submission to ApJL.

PRESENTATIONS

Selected Talks

TDEs and Nuclear Transients in Crete	2026
Inaugural Tinsley workshop	2024
LANL Center for Nonlinear Studies Student Talk Series (1st place student talk)	2024
KITP Program - Towards a Physical Understanding of Tidal Disruption Events	2024
RAMSES User Meeting	2024
HEAD Frontiers Seminar Series (invited)	2023
HEAD-19 Conference (invited)	2022

Selected Posters

AAS-245 Conference (Chambliss honorable mention)	2025
HEAD-20 Conference (1st place undergraduate poster)	2023
Connecticut Space Grant Consortium Expo	2021, 2022
Blue Waters Symposium for Petascale Science and Beyond	2018, 2019

SKILLS

Computer languages: PYTHON, C, C++, FORTRAN, HTML/CSS/JAVASCRIPT, Unix shell
 Software: H-AMR, RAMSES, ATHENA++/K, CLOUDY, MESA
 Languages: English (native), Spanish (conversational), Modern Greek (basic)
 Hobbies: Triathlon, jazz piano

GRANTS, FELLOWSHIPS, AND AWARDS

DOE Computational Science Graduate Fellowship, Krell Institute	2023 - 2027
Martin Schwarzschild Fellowship, Princeton University (departmental award)	2023 - 2025
Michael Manzella Award, Yale University (leadership award)	2023
Lamat Fellowship, University of California Santa Cruz (REU)	2022
Hahn Scholarship, Yale University (competitive undergrad research program)	2019 - 2021
Student Project Grant, NASA Connecticut Space Grant Consortium	2020
First-Year Summer Research Fellowship, Yale University	2020

LEADERSHIP EXPERIENCE

Thunch (thursday lunch) seminar series organizer	2024
Yale Undergraduate Aerospace Association; Project Leader, Director of Projects, President	2020 - 2023
Yale Club Triathlon, Captain	2021 - 2022

PROFESSIONAL SERVICE

Number of papers refereed
 MNRAS: 3
 YouTube channel with scientific visualizations
<https://www.youtube.com>

TEACHING, OUTREACH, AND SCIENCE COMMUNICATION

Recurring

Teaching fellow, Princeton University Preparatory Program	2025 - 2027
Mentored undergraduate researcher for summer research program	2026
Teaching assistant, AST 206: Black Holes	2024 - 2026
Tutor, New Jersey Prison Teaching Initiative	2024
Organizer, multiple aerospace-themed educational events with local schools	2021

One-time

Scientific visualization featured in Quanta magazine <i>Astrophysicists Puzzle Over Webb's New Universe</i> www.quantamagazine.org/...20260702	7/22, 2026
Volunteer, Princeton Spring into Science (educational event with local schools)	4/16, 2026
Speaker, Planetarium at New Jersey State Museum	2/21, 2026
Speaker, Astronomy on Tap (Trenton, NJ)	11/20, 2024
Speaker, solar eclipse at Littlebrook Elementary	4/08, 2024
Teacher, Yale Splash (annual event with local schools)	4/16, 2022